



QXU6033 – Advanced Polymer Chemistry

Lectures Notes 2: CRP: ATRP/SARA ATRP/ARGET/RAFT

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Textbooks:

“Polymers: Chemistry and Physics of Modern Materials” by J.M.G Cowie & Valeria Arrighi (CRC PRESS, 2008, 3rd Edition)

Also recommended for reference:

“Organic Chemistry” by Francis A. Carey & Robert M. Giuliano (10th Edition, McGraw-Hill Education, 2017)



Controlled radical polymerisation: ATRP/SARA/ARGET/ RAFT

Polymers: Chemistry and Physics of Modern Materials, JMG Cowie, 3rd Ed., Chapters 3 (p79), 4 (p109), 5 (p142)

Read also file on: Features of CRP (QMPLUS)

TOWARDS “LIVING” FRP: Kinetics issue!

• 3+1 Processes:

1- Initiation:

$[i^*]$, $[M]$ **No effect**

2- Propagation:

$[P^*]$, $[M]$ ↓

3- Termination:

2 order $[P^*]$ ↓

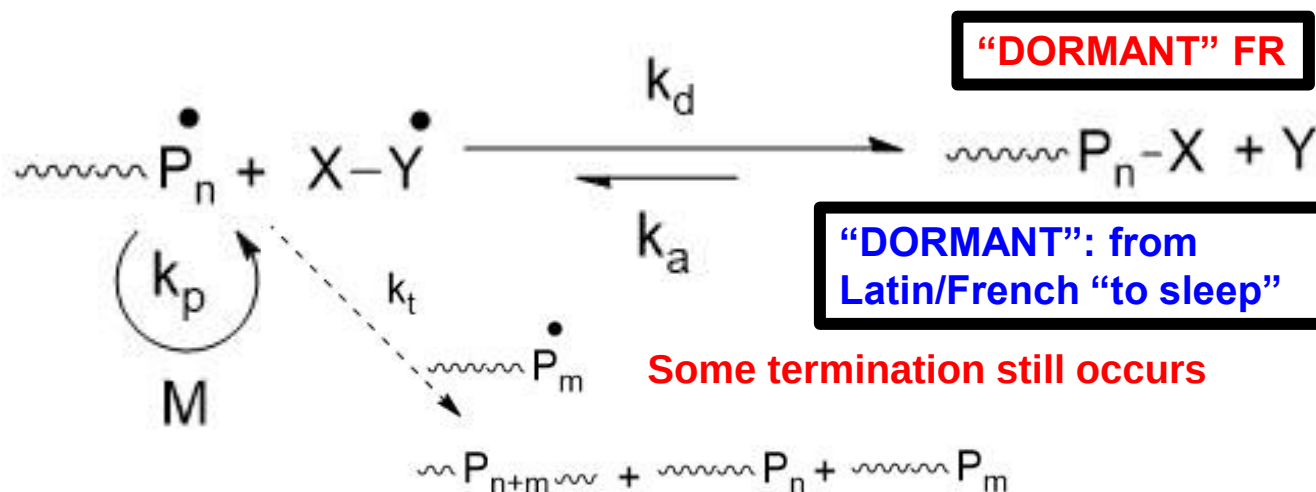
“+1”- Chain-Transfer:

$[P^*]$, $[CTA]$ ↓

Need to reduce $[P^*]$ but NOT TOTALLY

STRATEGY 2: “Assisted”/Catalysed REVERSIBLE TERMINATION

(“Catalytic REVERSIBLE DEACTIVATION”)



k_d : rate constant of “deactivation” (FR deactivation)

k_a : rate constant of “activation” (FR activation)

To be effective: $k_d > k_a$

STRATEGY 2: “Catalytic” REVERSIBLE DEACTIVATION: Through redox process, atom or group transfer (X can be a single atom transferred to P^* , Y normally a transition metal)

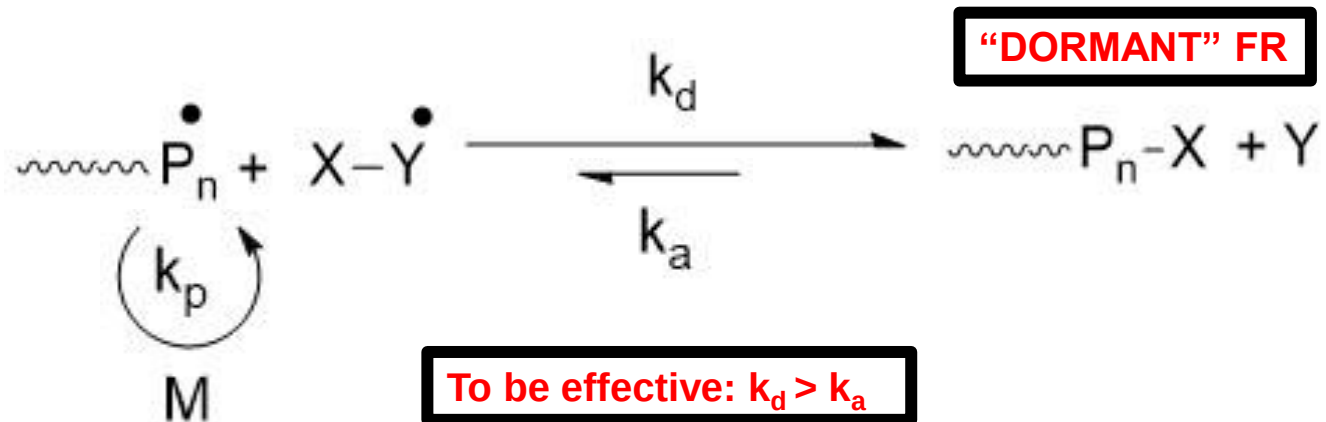
TOWARDS “LIVING” FRP: Kinetics issue!

• 3+1 Processes:

Need to reduce $[P^*]$ but NOT TOTALLY

STRATEGY 2: “Assisted”/Catalysed REVERSIBLE TERMINATION (“Catalytic REVERSIBLE DEACTIVATION”)

- 1- Initiation:
 $[i^*], [M]$ **No effect**
- 2- Propagation:
 $[P^*], [M]$ **↓**



- 3- Termination:
 2 order **$[P^*]$** **↓**
- “+1”- Chain-Transfer:
 $[P^*], [CTA]$ **↓**

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2- Propagation:

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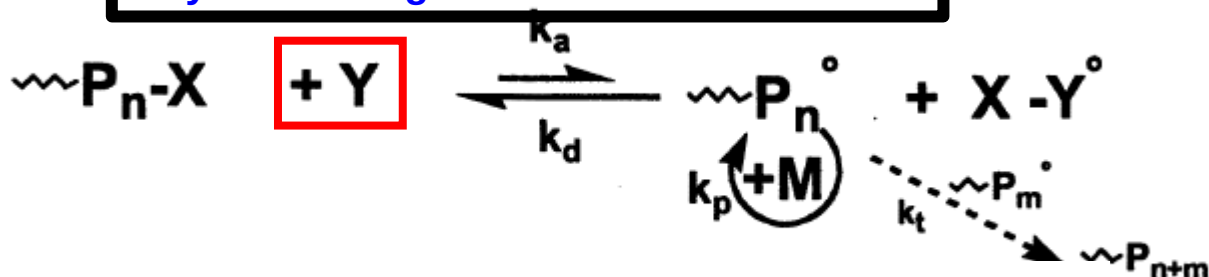
3- Termination:

2 order $[P^*]$ ↓

“+1”- Chain-Transfer:

$[P^*]$, $[CTA]$ ↓

This one is useful to look the opposite way at this stage



Providing a compound with a C-X group + Y will initiate the polymerisation (no need for heating or external FR initiators)

STRATEGY 2: “Catalytic” REVERSIBLE DEACTIVATION : Through redox process, atom or group transfer (X can be a single atom transferred to P^* , Y normally a transition metal)

CONTROLLED (Free) Radical POLYMERISATION:

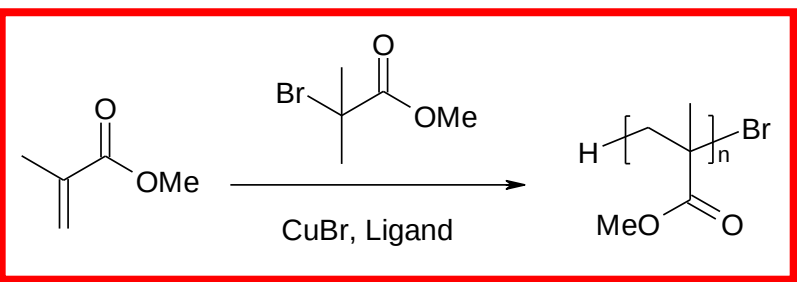
STRATEGY 2: “Assisted”/Catalysed REVERSIBLE TERMINATION

(“Catalytic REVERSIBLE DEACTIVATION”)

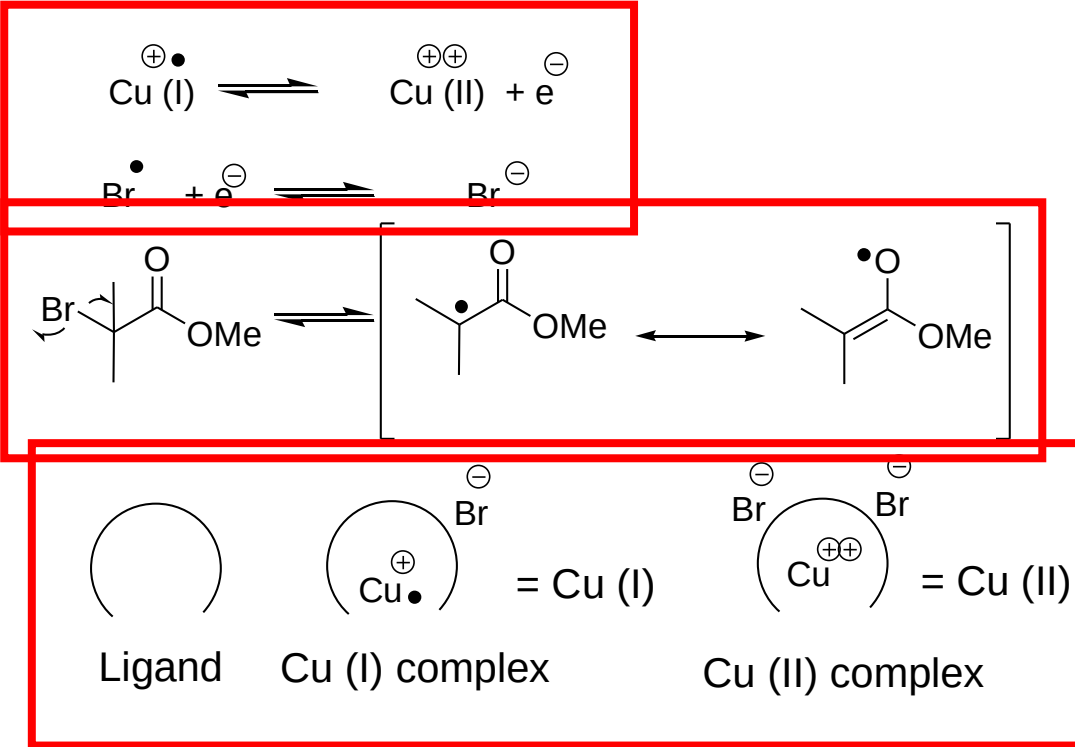
(Redox/Atom/Group Transfer)

Specific chemistries:

ATRP (Atom-transfer free radical polymerisation). A bromo (or chloro) compound + Cu^+ complex act as the initiator: does not require an external free-radical initiation system



Series of redox processes (metal mediated exchange of 1 e, or free radical), bromo-carbon species serve as “dormant” free radicals. Works well with Cl as well, instead of Br.

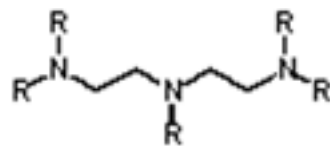


FROM NOW ON: DO NOT USE the term “Strategy 2”: refer to specific chemistry (ATRP)

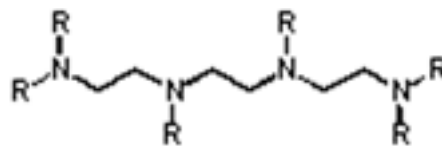
CONTROLLED (Free) Radical POLYMERISATION:



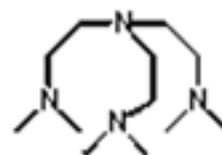
Typical ligands: chelating “polyamines” (poly from inorganic chemistry concept of more than 1, rather than polymer)



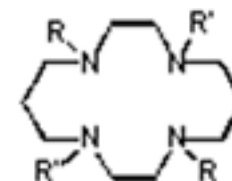
DETA (R = H)
PMDETA (R = Me)



TETA (R = H)
HMTETA (R = Me)



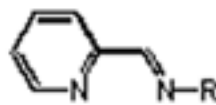
Me₆TREN



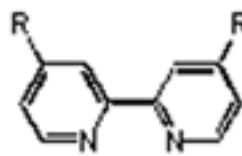
cyclam (R = R' = R'' = H)
Me₄Cyclam (R = R' = R'' = Me)
DMCBCy (R = Me, R' + R'' = CH₂CH₂)



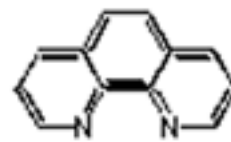
Gln-R



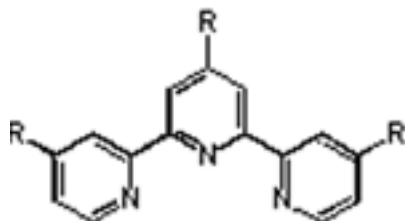
Pym-R



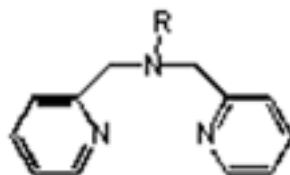
bpy (R = H)
dNbpy (R = CHBu₂)



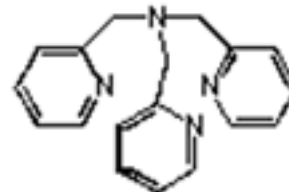
1,10-Phen



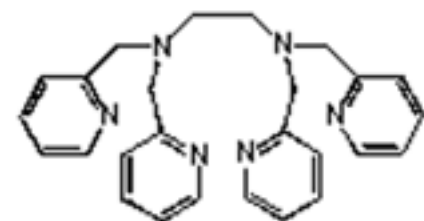
tpy (R = H)
tNbpy (R = CHBu₂)



BPMPra (R = Pr)
BPMA (R = C₂H₁₇)
BPMDA (R = C₁₂H₂₇)



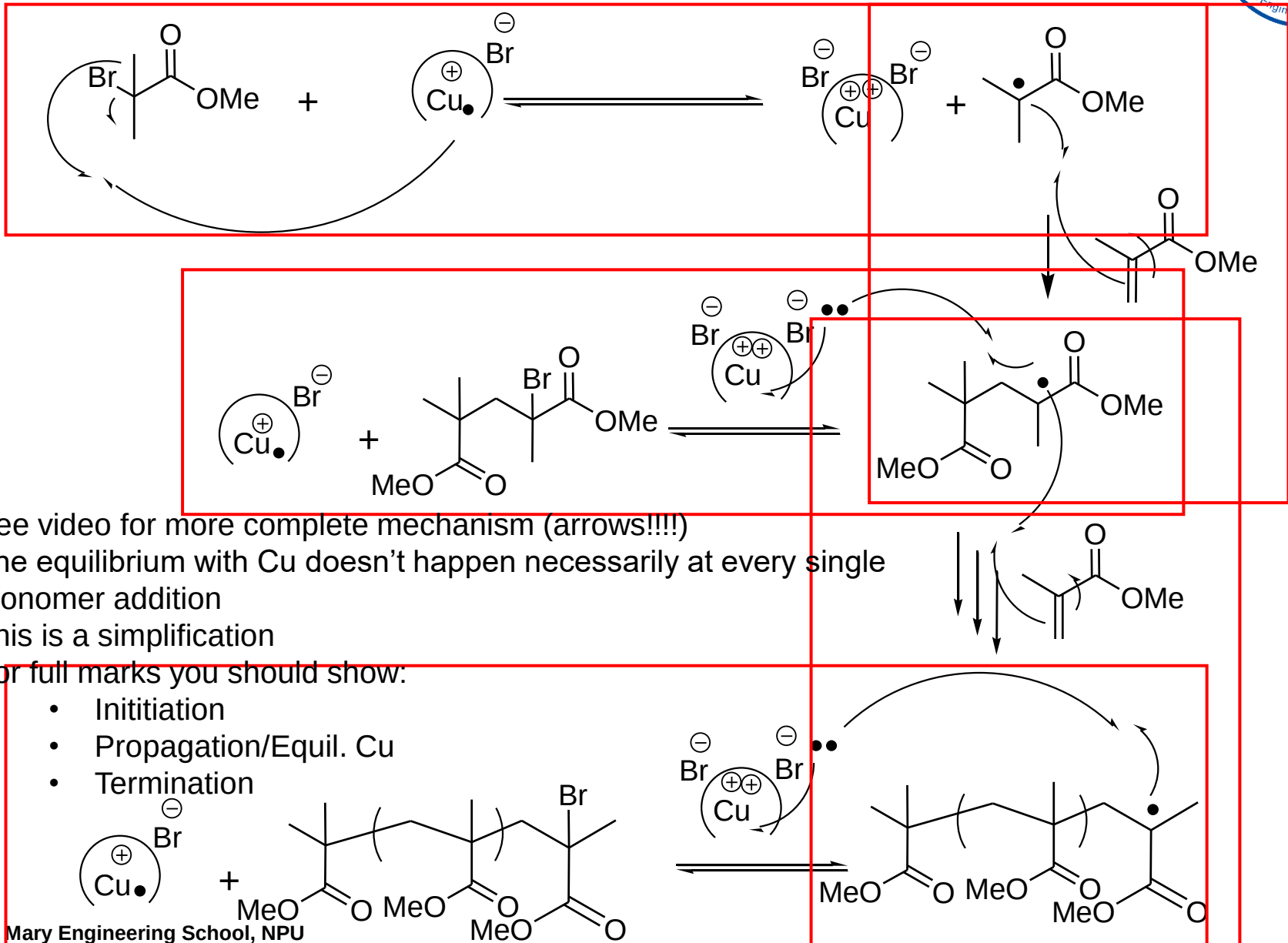
TPMA



TPEDA

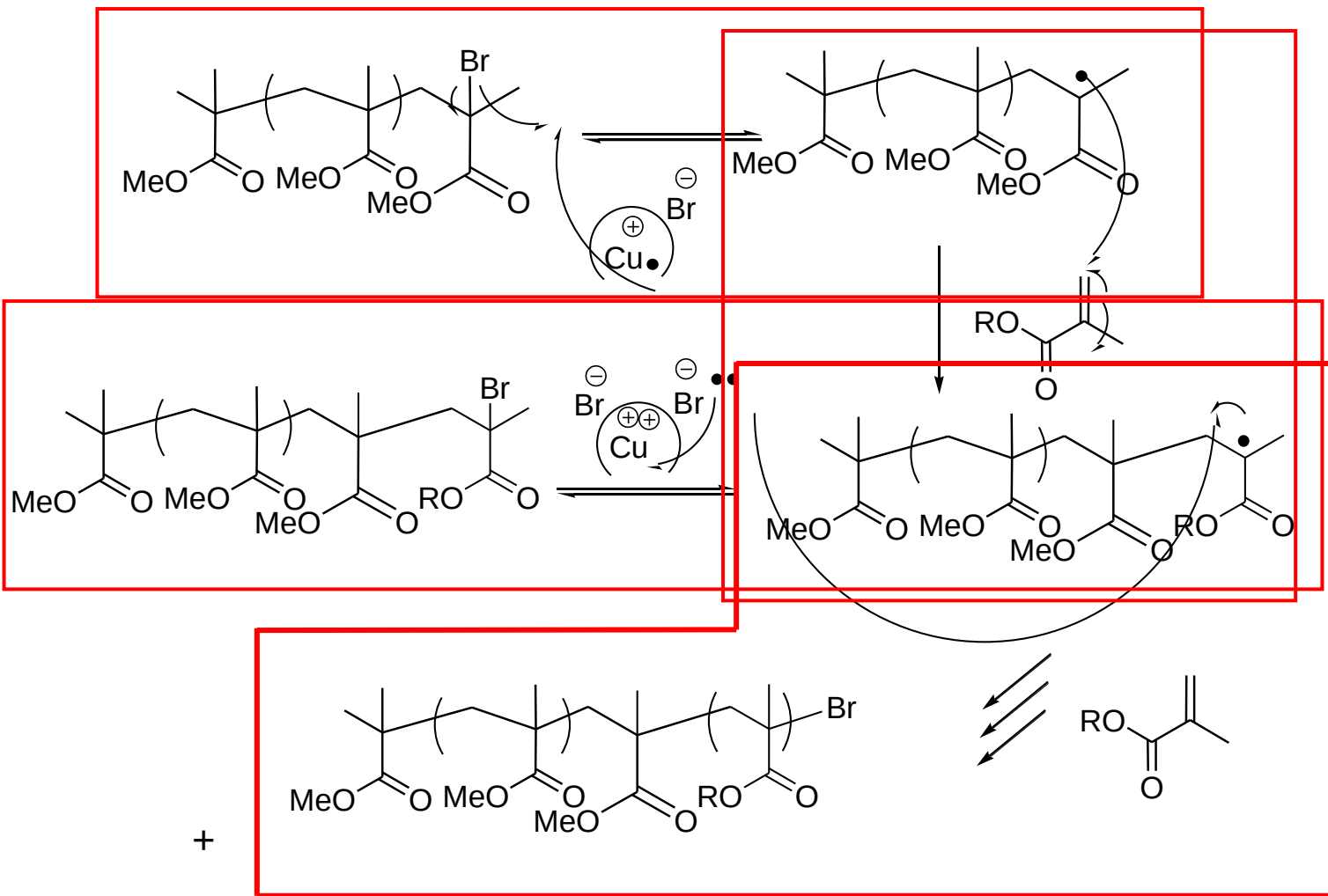
CONTROLLED (Free) Radical POLYMERISATION:

ATRP (Atom-transfer free radical polymerisation)



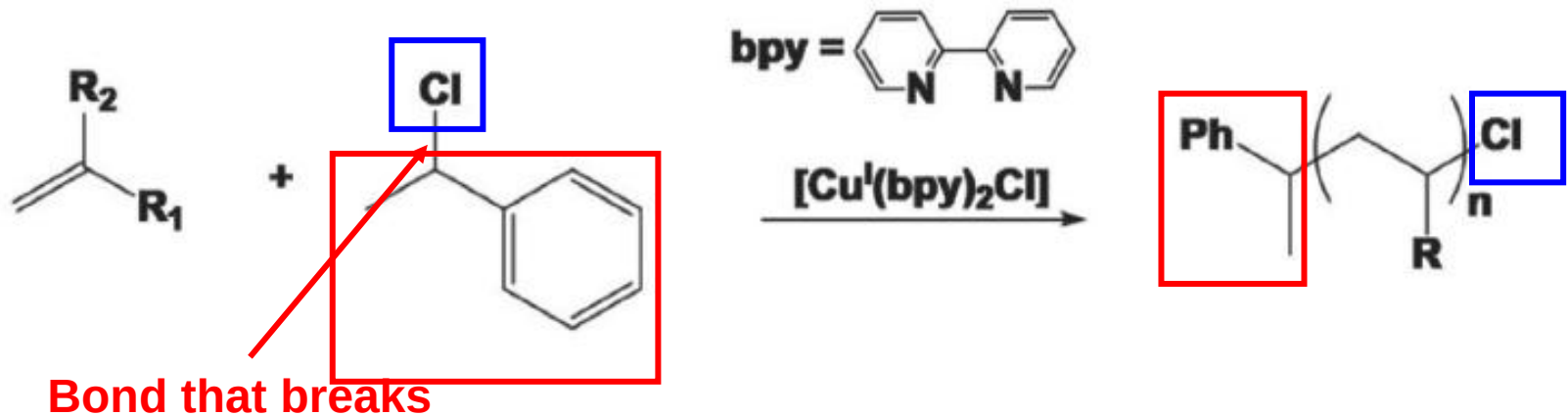
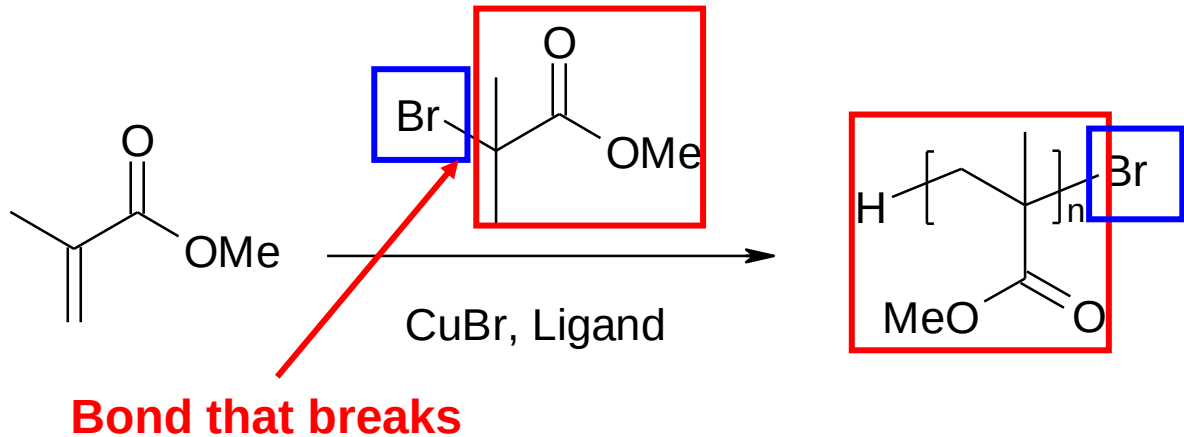
CONTROLLED (Free) Radical POLYMERISATION:

ATRP: Like NMP: you can isolate the Br (or Cl) terminated polymer, add more monomer and Cu complex and synthesise another block



CONTROLLED (Free) Radical POLYMERISATION:

ATRP: END-GROUPS: DEPEND ON INITIATOR: POLYMER GROWS IN BETWEEN THE BONDS OF THE INITIATOR THAT BREAKS



CONTROLLED (Free) Radical POLYMERISATION:

ATRP:

Advantages:

- Simple
- No need to heat up
- Tolerant of a range of functional groups

Disadvantages:

- Difficult to replicate literature results
- Very sensitive to O_2 (will oxidise $Cu^+ \rightarrow Cu^{++}$, stopping activation)
- Amino/amide/ $COOH$ -containing monomers compete for ligand and change kinetics (usually not favourably)
- Cu is toxic (removal of Cu from biomaterials is difficult)

CONTROLLED (Free) Radical POLYMERISATION:

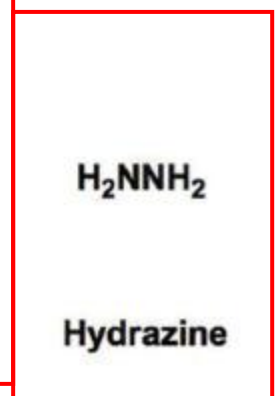
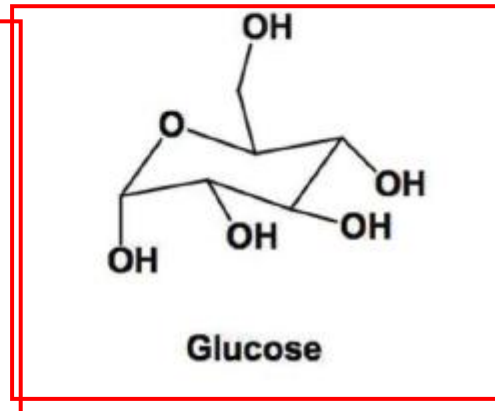
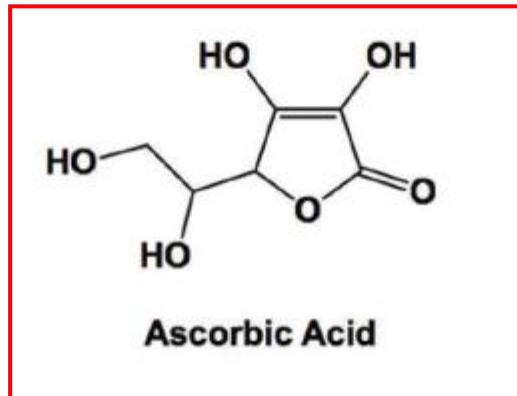
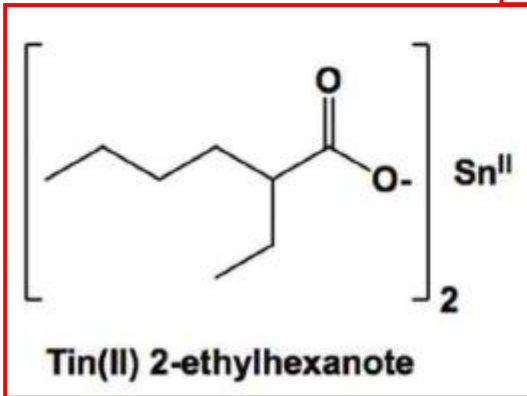
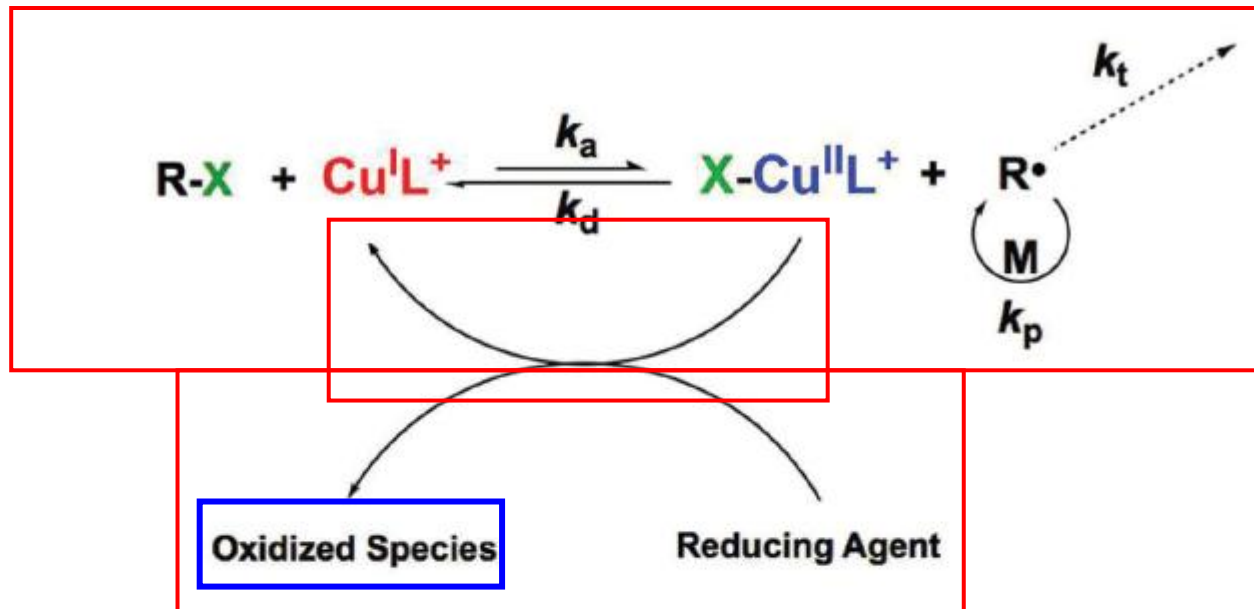
To improve ATRP: a range of methods have been devised:
(here only a few examples are shown)

ARGET-ATRP: Activators ReGenerated by Electron Transfer

- Addition of external reducing agents to convert $\text{Cu}^{++} \rightarrow \text{Cu}^{+}$
- Lower quantity of Cu needed (this is good, as Cu can be toxic if biomaterials are prepared)
- More tolerant of O_2
- Reducing agents used: tin(II) 2-ethylhexanoate ($\text{Sn}(\text{EH})_2$), glucose, ascorbic acid, silver metal, hydrazine ($\text{H}_2\text{N}-\text{NH}_2$) or phenyl hydrazine ($\text{Ph}-\text{NH}-\text{NH}_2$) ($\text{C}_6\text{H}_5-\text{NH}-\text{NH}_2$).

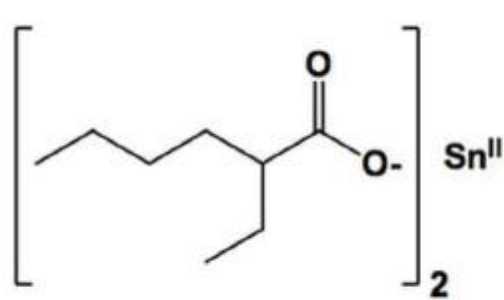
CONTROLLED (Free) Radical POLYMERISATION:

ARGET-ATRP: Activators ReGenerated by Electron Transfer



CONTROLLED (Free) Radical POLYMERISATION:

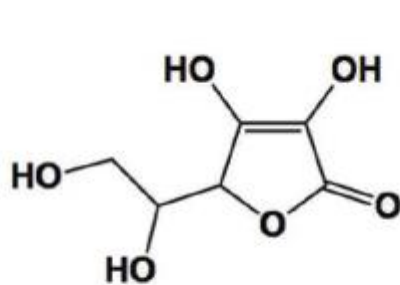
ARGET-ATRP: Activators ReGenerated by Electron Transfer



Tin(II) 2-ethylhexanoate

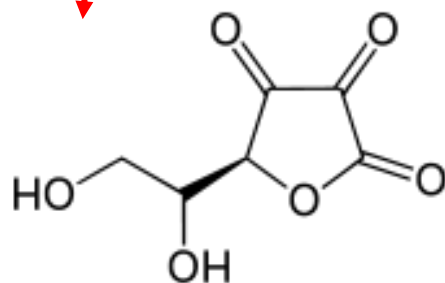


Sn^{IV}

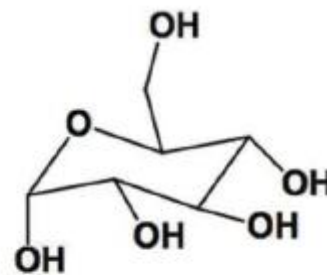


Ascorbic Acid

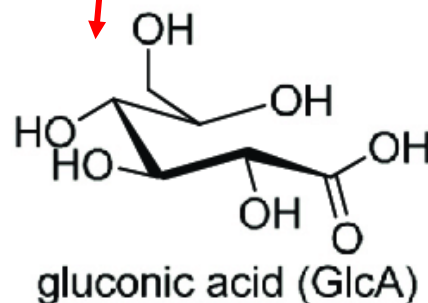
Oxidized Species



Dehydroascorbic acid



Glucose



gluconic acid (GlcA)



Hydrazine



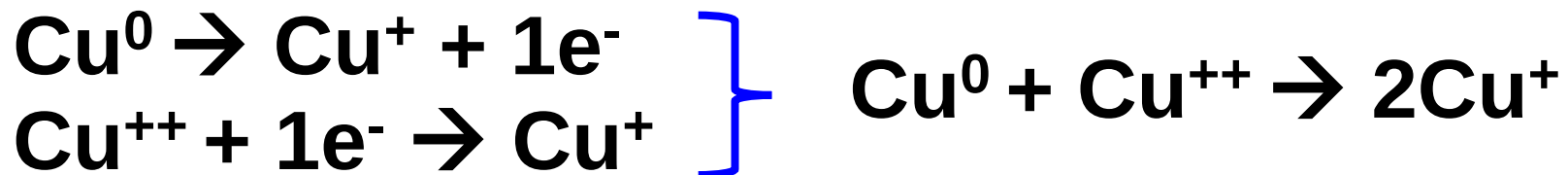
N_2

CONTROLLED (Free) Radical POLYMERISATION:

To improve ATRP: a range of methods have been devised:
(here only a few examples are shown)

SARA-ATRP: Supplemental Activation Reducing Agent:

- Addition of Cu^0 to both reduce $\text{Cu}^{++} \rightarrow \text{Cu}^+$ (major role) AND activate C-Br/C-Cl to C^* (FR) (minor role).
- More tolerant of O_2
- In the literature, you may also find SET-LRP (Single Electron Transfer-Living Free Radical Polymerisation)
- SET-LRP is proposed as an alternative mechanism for SARA-ATRP, where Cu^0 mostly acts as activator of C-Br/C-Cl to C^*
- Experimental evidence seems to support SARA-ATRP as the correct mechanism (ATRP occurs “normally”, with Cu^0 regenerating the main activator, Cu^+)
- The system is complex, solvent dependent/concentration dependent.



TOWARDS “LIVING” FRP: Kinetics issue!

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$[i^*]$, $[M]$ **No effect**

2- Propagation:

$[P^*]$, $[M]$ ↓

3- Termination:

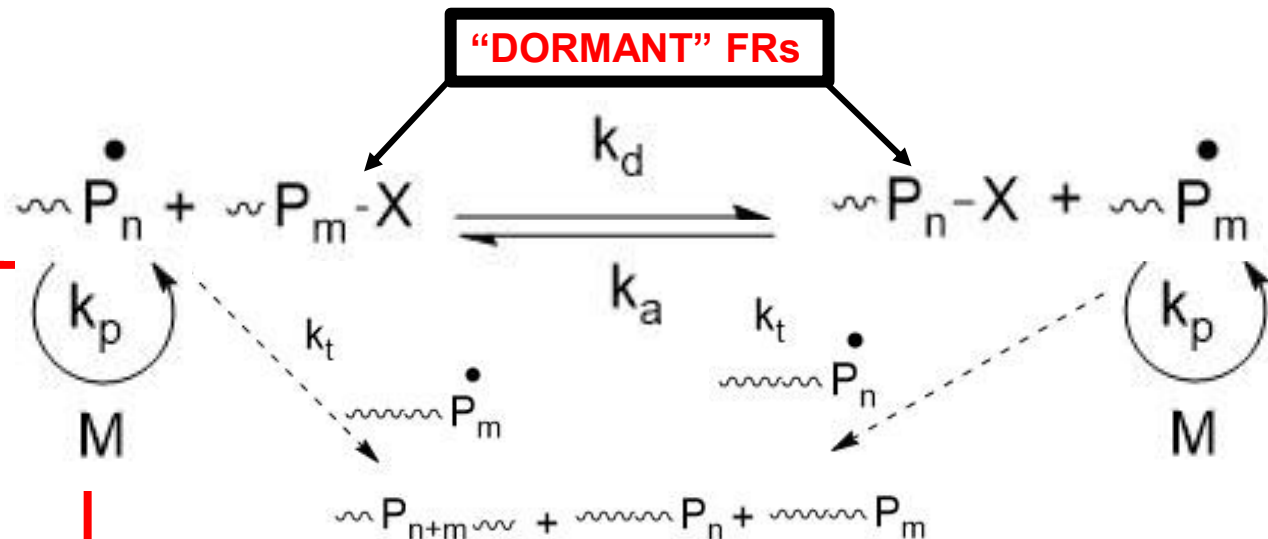
2 order $[P^*]$ ↓

“+1”- Chain-Transfer:

$[P^*]$, $[CTA]$ ↓

Need to reduce $[P^*]$ but NOT TOTALLY

STRATEGY 3: REVERSIBLE CHAIN-TRANSFER
 (“DEGENERATIVE TRANSFER”)



Some termination still occurs

k_d : rate constant of “deactivation” (FR deactivation)

k_a : rate constant of “activation” (FR activation)

$$k_d \sim k_a$$

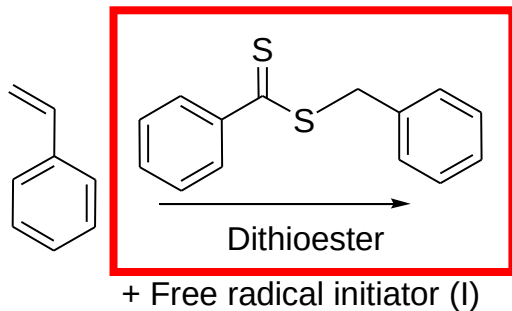
CONTROLLED (Free) Radical POLYMERISATION:

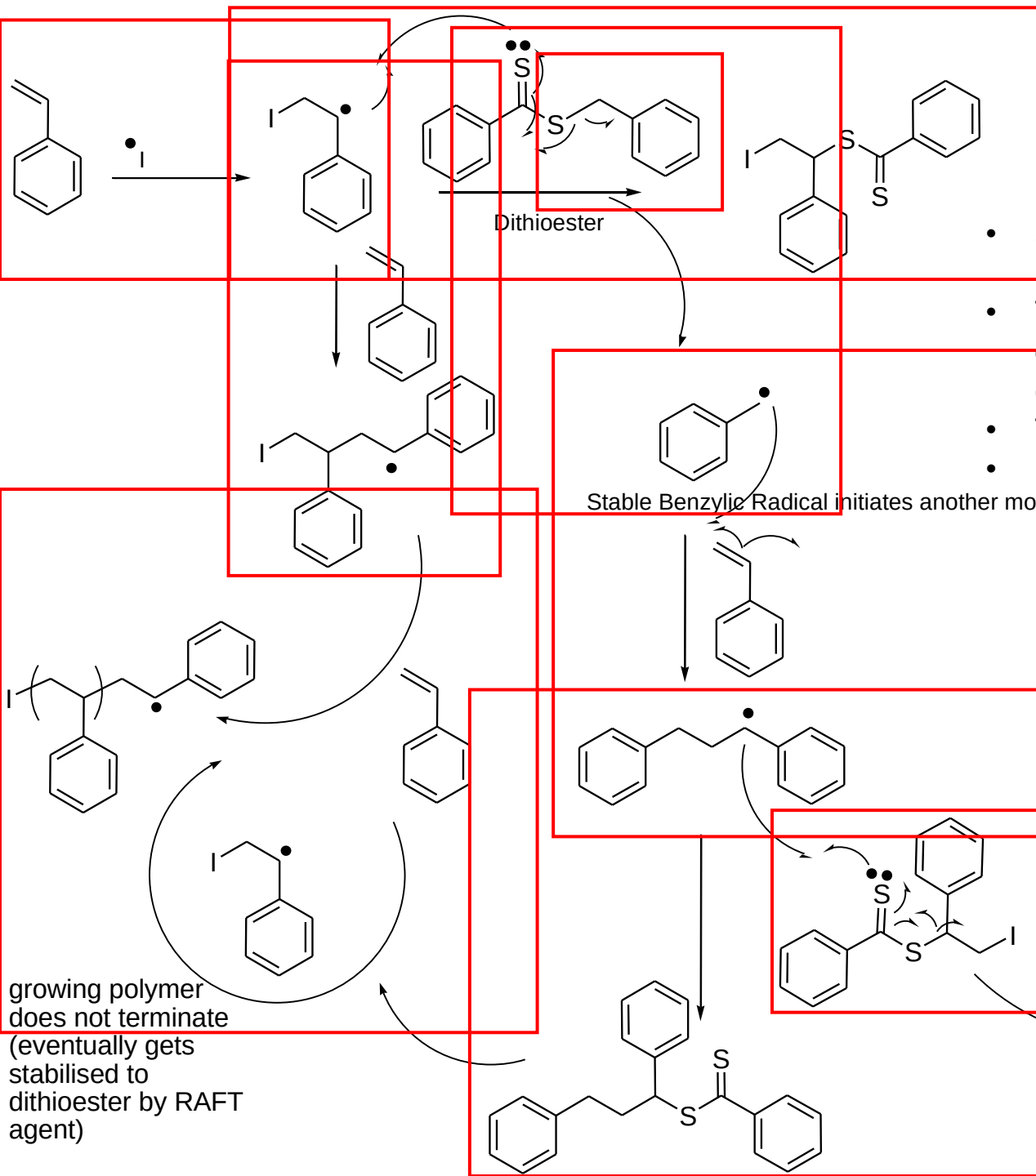
STRATEGY 3: REVERSIBLE CHAIN-TRANSFER (DEGENERATIVE TRANSFER)

Specific chemistries:

RAFT: Reversible addition-fragmentation chain transfer polymerisation:

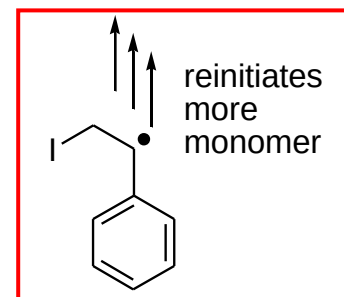
Requires an external free radical initiator and a RAFT agent (dithioester, or similar species)



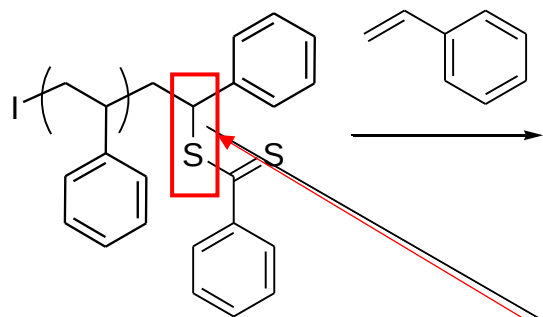


- See video for more complete mechanism (arrows!!!!)
- The reaction with dithioester doesn't happen necessarily at every single monomer addition
- This is a simplification
- For full marks you should show:

- Initiation
- Propagation/Equil. dithioester
- Termination

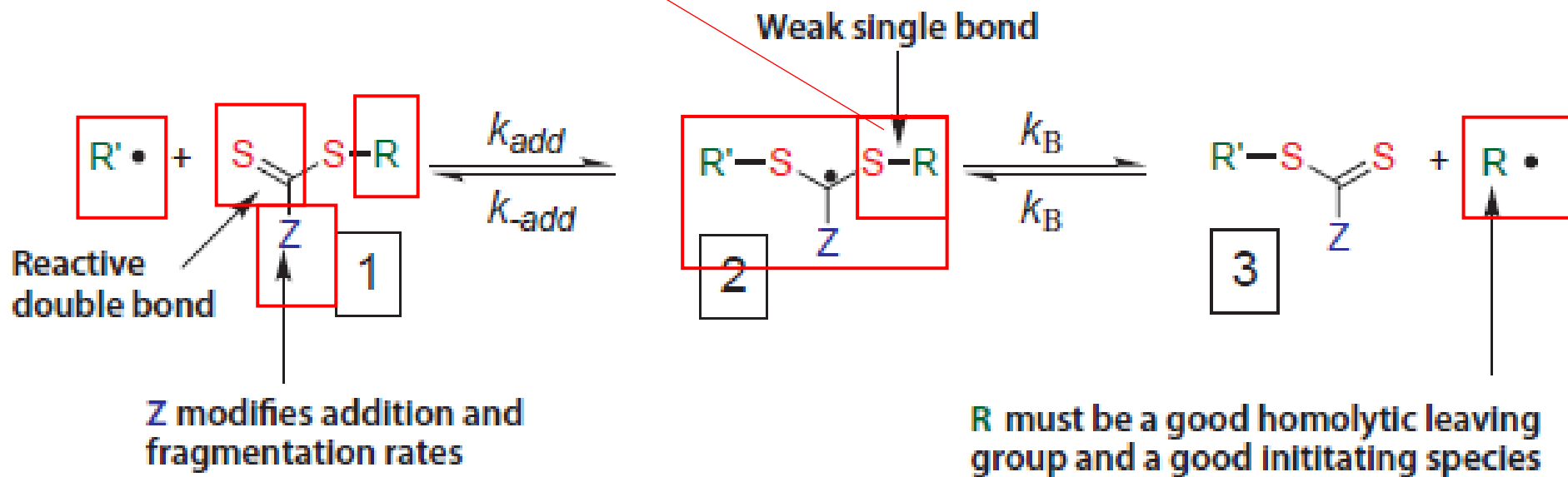


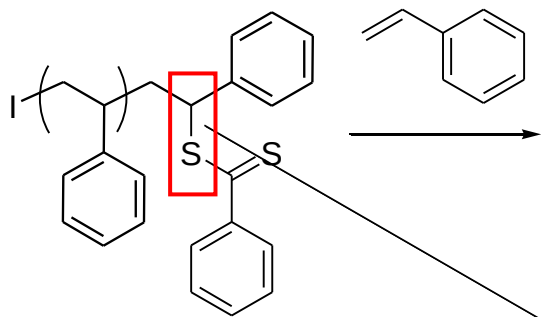
growing polymer does not terminate (eventually gets stabilised to dithioester by RAFT agent)



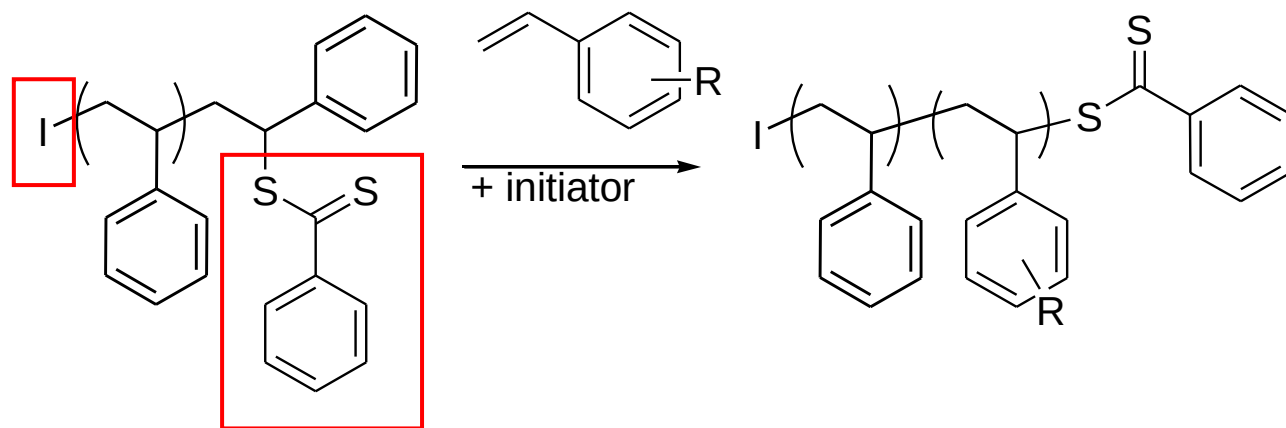
In summary:

Effectively: Progressively more monomer gets "inserted" between this C-S bond.

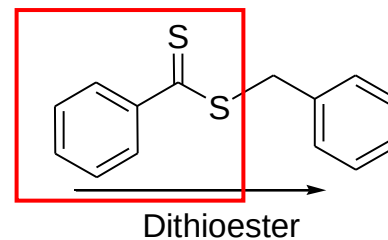




Effectively: Progressively more monomer gets "inserted" between this C-S bond.



**End-groups:
depend on
initiator and
RAFT AGENT**



After isolation of dithioester-terminated polymer:
A second monomer can be added, + initiator to synthesise a block copolymer

RAFT Agents

Weak single bond

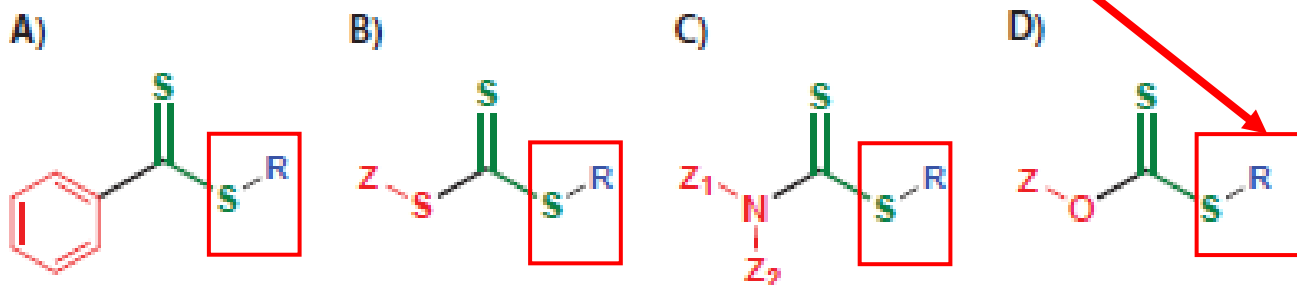


Figure 3. Four classes of RAFT agents: **A)** Dithiobenzoates, **B)** Trithiocarbonates, **C)** Dithiocarbamates, and **D)** Xanthates.

A) Dithiobenzoates

- Very high transfer constants
- Prone to hydrolysis
- May cause polymerization retardation under high concentration

B) Trithiocarbonates

- High transfer constants
- More hydrolytically stable (than dithiobenzoates)
- Cause less retardation

C) Dithiocarbamates

- Activity determined by substituents on N
- Effective with electron-rich monomers

D) Xanthates

- Lower transfer constants
- More effective with less activated monomers
- Made more active by electron-withdrawing substituents

RAFT Agents

Weak single bond

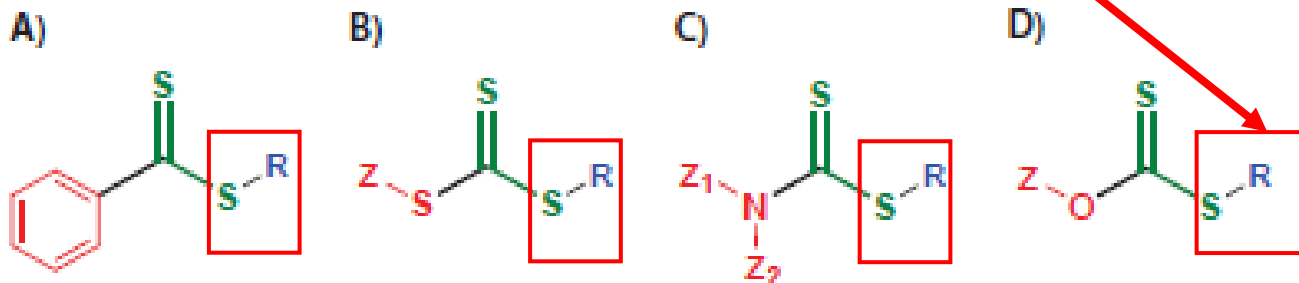
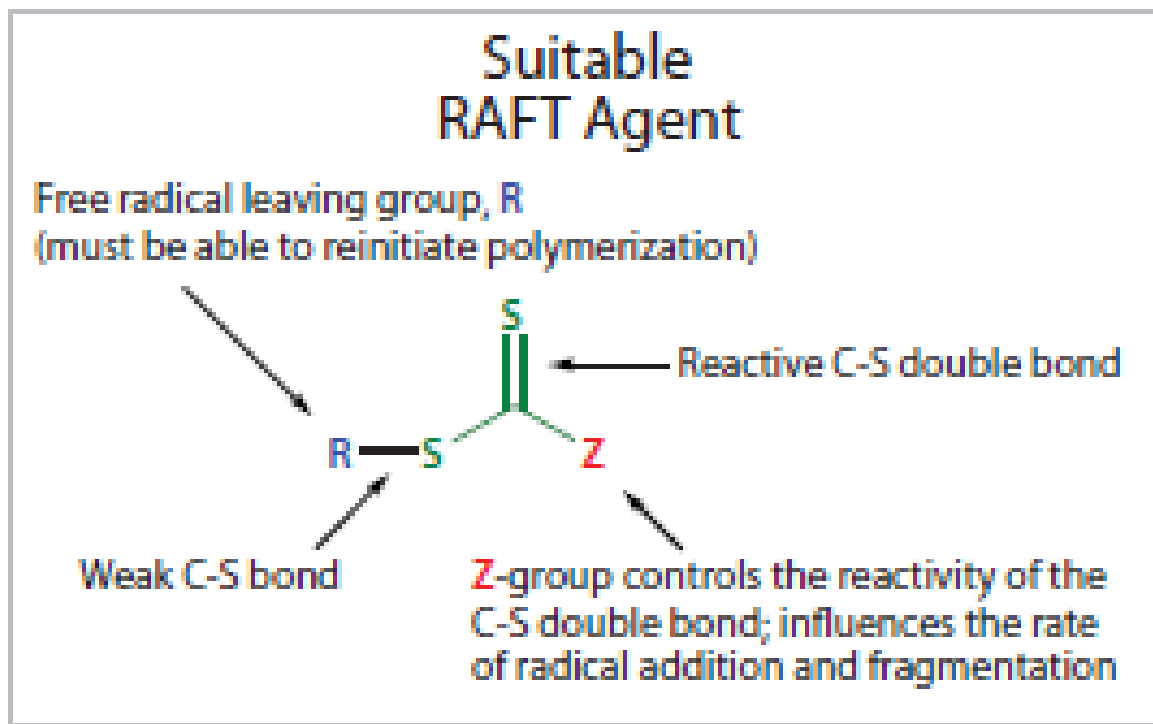


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<https://www.sigmaaldrich.com/content/dam/sigma-aldrich/docs/SAJ/Brochure/1/controlled-radical-polymerization-guide.pdf>

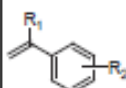
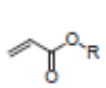
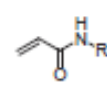
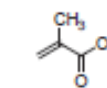
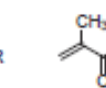
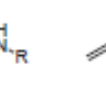
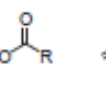
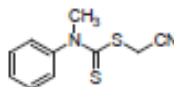
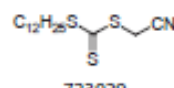
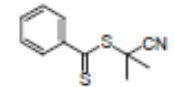
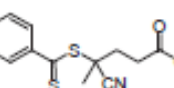
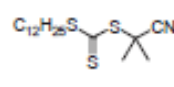
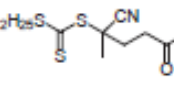
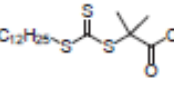


RAFT Agents

<https://www.sigmaaldrich.com/content/dam/sigma-aldrich/docs/SAJ/Brochure/1/controlled-radical-polymerization-guide.pdf>

For a complete list see also (and original document)

https://www.sigmaaldrich.com/materials-science/material-science-products.html?TablePage=103936134&utm_source=redirect&utm_medium=promotional&utm_campaign=raftagent

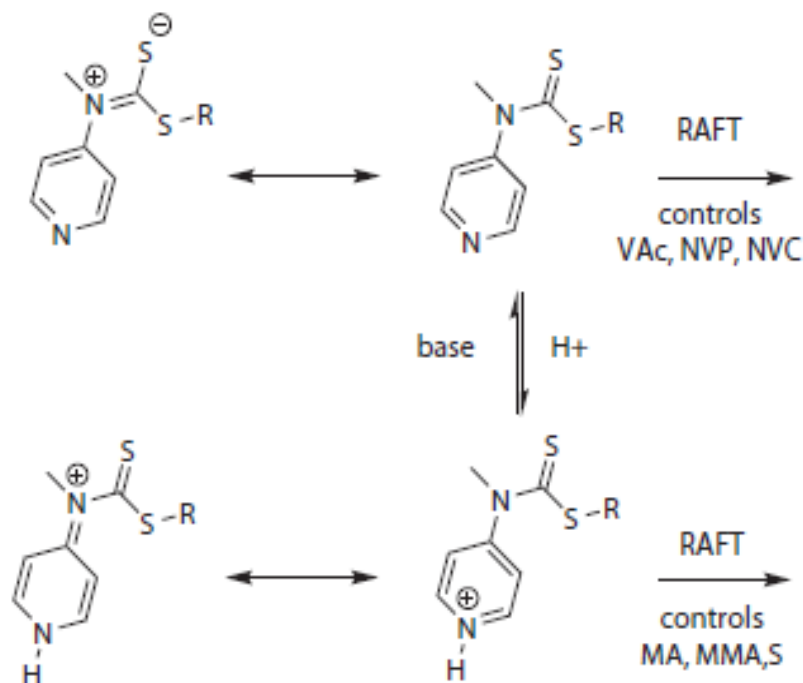
	 styrenes	 acrylates	 acrylamides	 methacrylates	 methacrylamides	 vinyl esters	 vinyl amides
 723002	-	-	-	-	-	+++	+++
 723029	+++	+++	+++	-	-	-	-
 722987	++	+	-	+++	+++	-	-
 722995	++	+	+	+++	+++	-	-
 723037	+++	++	++	+++	+++	-	-
 723274	+++	++	++	+++	+++	-	-
 723010	+++	+++	+++	+	+	-	-

+++ = Well-suited ++ = Moderately suited + = Variable results - = Poorly suited

Switchable-RAFT Agents

<https://www.sigmaaldrich.com/content/dam/sigma-aldrich/docs/SAJ/Brochure/1/controlled-radical-polymerization-guide.pdf>

MAMs: more activated monomers, LAMs: less activated monomers



Sty: Styrene
MA: Methyl acrylate
MMA: MMA
VAc: Vinyl acetate
NVP: N-Vinyl pyrrolidone
NVC: N-Vinyl carbazole

Scheme 1. Differences in the canonical forms of protonated and deprotonated *N*-methyl, *N*-(4-pyridyl)dithiocarbamates. When protonated the agents can control polymerization of MAMs (such as Sty, MA, MMA) and then when deprotonated, the RAFT agents can effectively control polymerization of LAMs (such as VAc, NVP and NVC).^{1,9}

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MAMs: more activated monomers, **LAMs:** less activated monomers

This way you can prepare block copolymers of MAMs and LAMs (by changing the pH when you change the monomer)

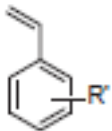
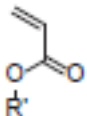
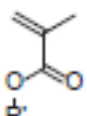
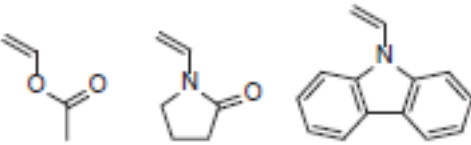
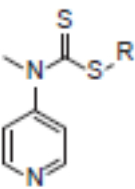
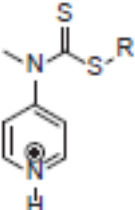
	 Styrenes	 Acrylates	 Methacrylates	 Vinyl esters/amides
	-	-	-	✓
	✓	✓	✓	-

Table 1. The neutral and protonated switchable RAFT agents with their suitability for selected monomer types (✓ = compatible; - = incompatible).

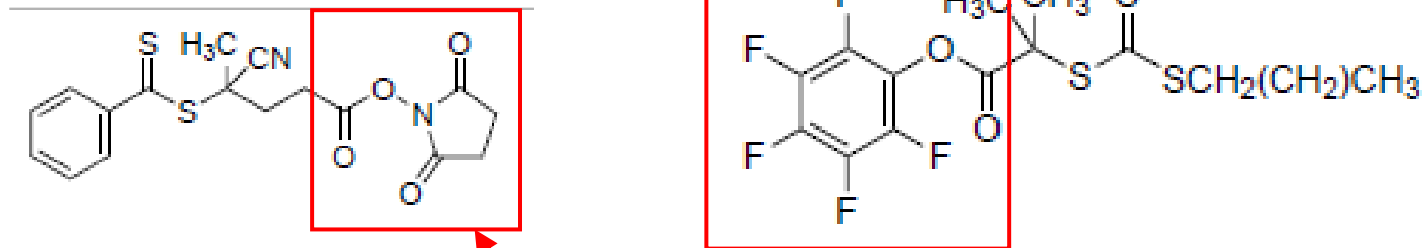
Functional-RAFT Agents

<https://www.sigmaaldrich.com/content/dam/sigma-aldrich/docs/SAJ/Brochure/1/controlled-radical-polymerization-guide.pdf>

RAFT agents that allow it to react with a functional group **BEFORE** the polymerisation

This way you can grow a polymer from another compound, including another polymer prepared by a different polymerisation process. A good way to prepare block copolymers.

Some examples:

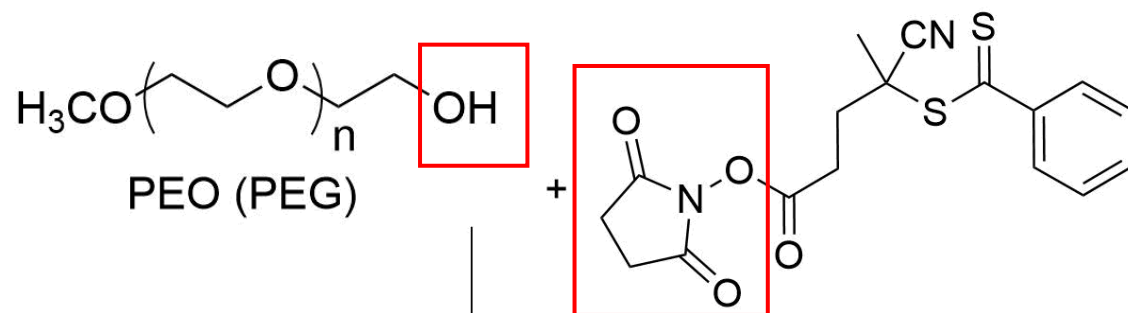


Activated esters

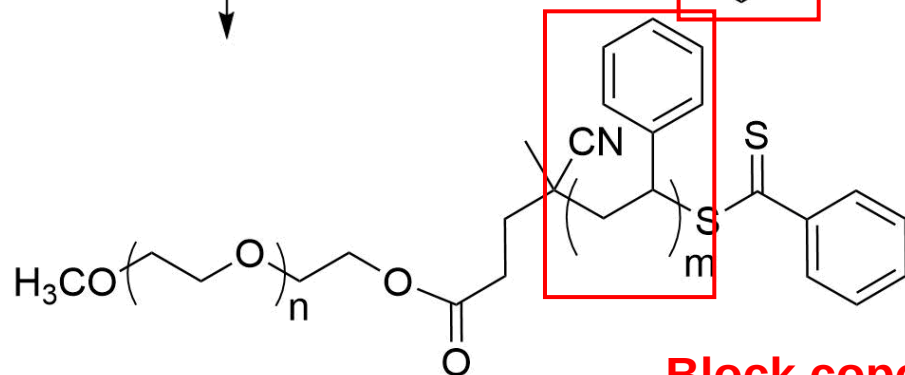
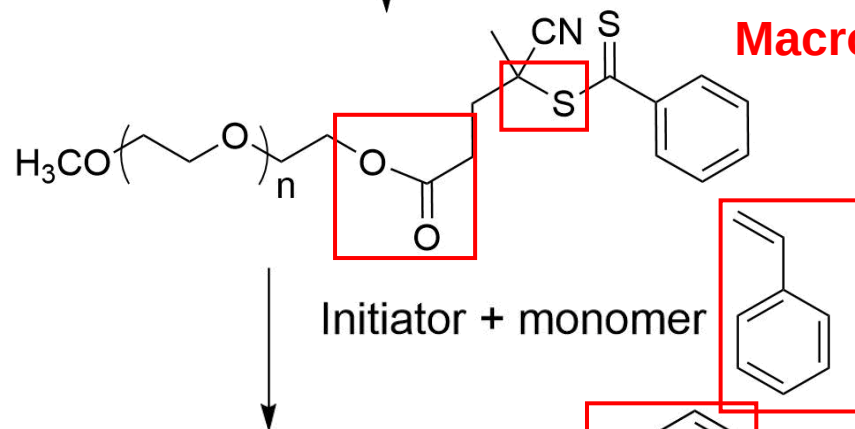
In an activated ester, the non-carbonyl oxygen is attached to a VERY electron-withdrawing group making it an excellent leaving group. Activated esters are as reactive as acid chlorides, but do NOT produce HCl as a by product.

Functional-RAFT Agents

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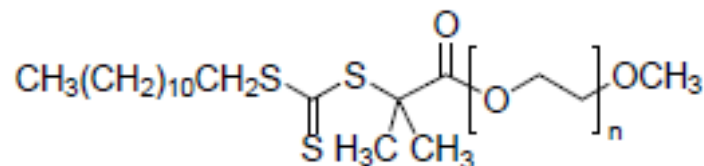
Macro-RAFT Agent (polymeric RAFT agent)



Block copolymer PEG-Styrene

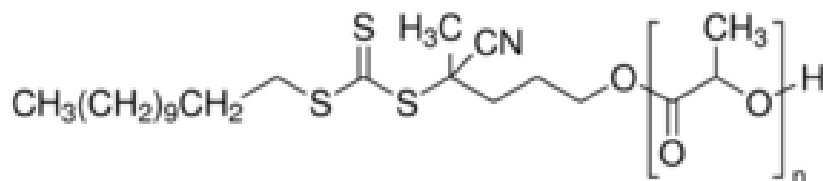
Commercially available Macro-RAFT Agents

<https://www.sigmaaldrich.com/content/dam/sigma-aldrich/docs/SAJ/Brochure/1/controlled-radical-polymerization-guide.pdf>



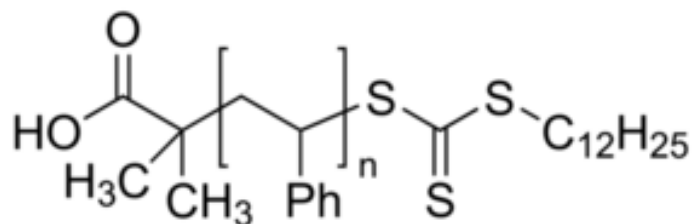
Poly(ethylene glycol) methyl ether (2-methyl-2-propionic acid dodecyl trithiocarbonate), average M_n 10,400

For PEG Block copolymers



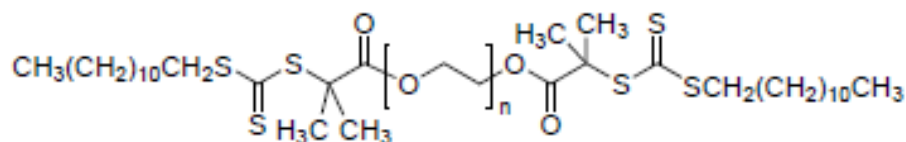
Poly(D,L-lactide), 4-cyano-4-[(dodecylsulfanylthiocarbonyl)sulfanyl]pentonate terminated average M_n 5000, PDI ≤ 1.5

For PLA Block copolymers



Polystyrene, DDMAT terminated average M_n 10,000, PDI ≤ 1.1

For Sty Block copolymers



Poly(ethylene glycol) bis[2-(dodecylthiocarbonothioylthio)-2-methylpropionate], average M_n 10,800

For PEG TRI-Block copolymers



CONTROLLED (Free) Radical POLYMERISATION:

FEATURES of CRP

Please see pdf on QMPLUS! (<https://www.cmu.edu/maty/crp/features.html>)

Ideal CRP will have:

- **First order kinetics (with regards to $[M]$)**
- **Controllable (can be designed) DP (in some papers noted as X)**
- **Controllable/narrow MW distribution ($PDI < 1.2-1.3$)**
- **Control of end-groups**

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- Control of end-groups

- **First order kinetics (with regards to $[M]$)**

Critical to experimentally demonstrate that CRP is working well (and that a system is indeed CRP)

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Ideal CRP will have: First order kinetics (with regards to [M])

- $R_p = -\frac{d[M]}{dt} = k_p[P^*][M]$
- Integrated: $\ln \frac{[M]_0}{[M]} = k_p[P^*]t$
- if $[P^*]=\text{constant}$: $\ln \frac{[M]_0}{[M]} = k'_p t$ (where $k'_p = k_p[P^*]$)
- R_p : rate of polymerisation; $[M]$: monomer concentration; k_p : rate constant of propagation; $[P^*]$: concentration of propagating (polymeric) free-radicals; $[M]_0$: original/starting monomer concentration

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Ideal CRP will have: First order kinetics (with regards to [M])

- $\overline{DP}_n = \frac{M_n}{M_0} = \frac{\Delta[M]}{[i]_0} = \frac{[M]_0 - [M]}{[i]_0} = \frac{[M]_0}{[i]_0} p$

- $PDI = \frac{\overline{DP}_w}{\overline{DP}_n} = \frac{M_w}{M_n} = 1 + \frac{\overline{DP}_n}{(\overline{DP}_n + 1)^2}$;

- note that at high DP: $PDI \approx 1 + \frac{1}{\overline{DP}_n}$

- $\overline{DP}_n$: number average degree of polymerisation (average number of repeating units in a polymer of MW= M_n (in previous modules you have seen this as simply DP); in some papers $\overline{DP}_n$ appears as $\overline{X}_n$ or X_n ; M_n : number average molecular weight; M_0 : monomer molar mass (PLEASE NOTE that M_0 is NOT $[M]_0$); $[i]_0$: original/starting initiator concentration; p : monomer conversion (fraction 0→1); PDI: dispersity/polydispersity index, M_w : weight average molecular weight, $\overline{DP}_w$: weight average degree of polymerisation (average number of repeating units in a polymer of MW= M_w)

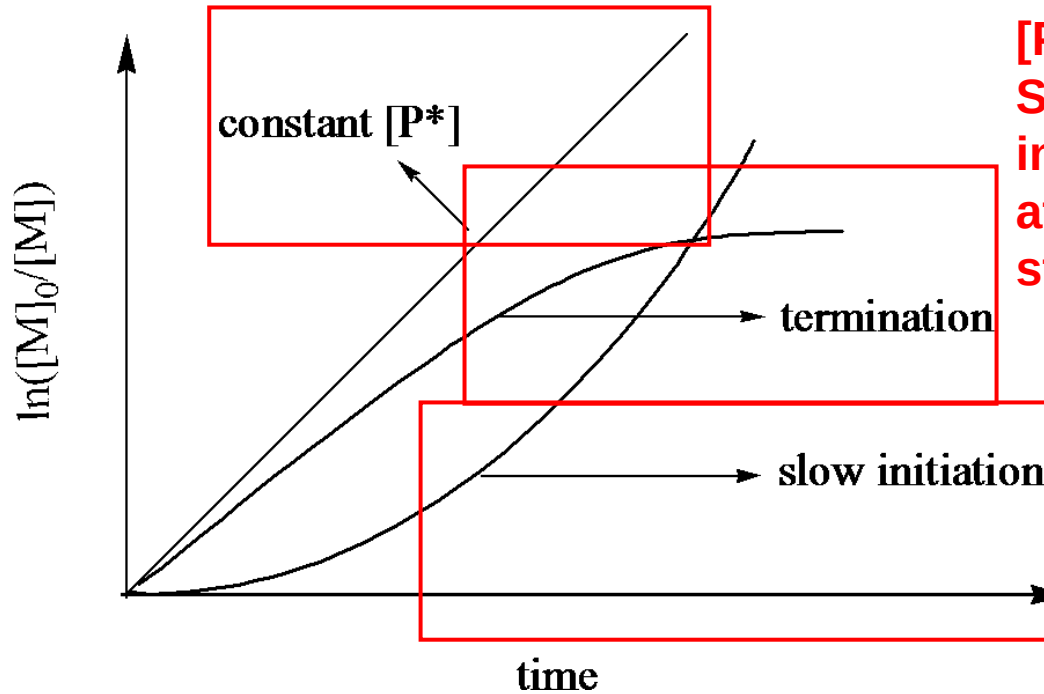
Note: you will see the use DP or $\bar{v}$ because we assume there is no termination!

CONTROLLED (Free) Radical POLYMERISATION:

FEATURES of CRP

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Ideal CRP will have: First order kinetics (with regards to $[M]$): how to demonstrate experimentally?: straight line $\ln[M]_0/[M]$ vs time (negative slope) or $\ln[M]_0/[M]$ vs time



$[P^*]$ might not be constant. Shape of curve gives good indication of slow initiation, or at which point termination may start.

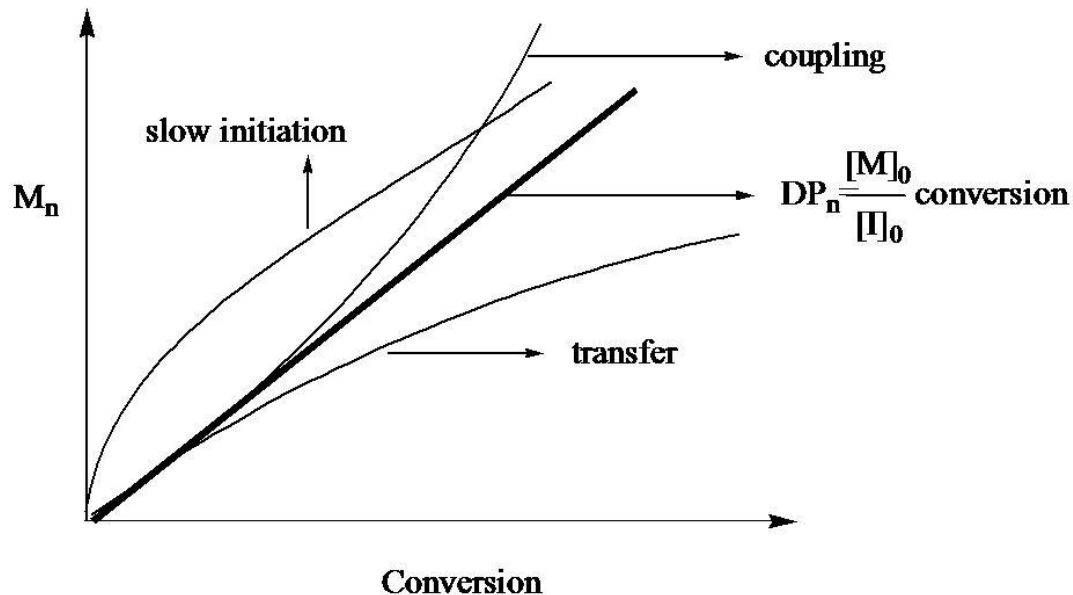
Termination: you can run the polymerisation up to the time in which termination will start dominating (therefore no longer controlled)

CONTROLLED (Free) Radical POLYMERISATION:

FEATURES of CRP

Please see pdf on QMPLUS! (<https://www.cmu.edu/maty/crp/features.html>)

Ideal CRP will have: First order kinetics (with regards to [M]): how to demonstrate experimentally? Straight line M_n vs p (conversion) This is a lot more common in the literature (M_n is relatively easy to measure).



Deviations from linearity give a good indication of what the problem may be.